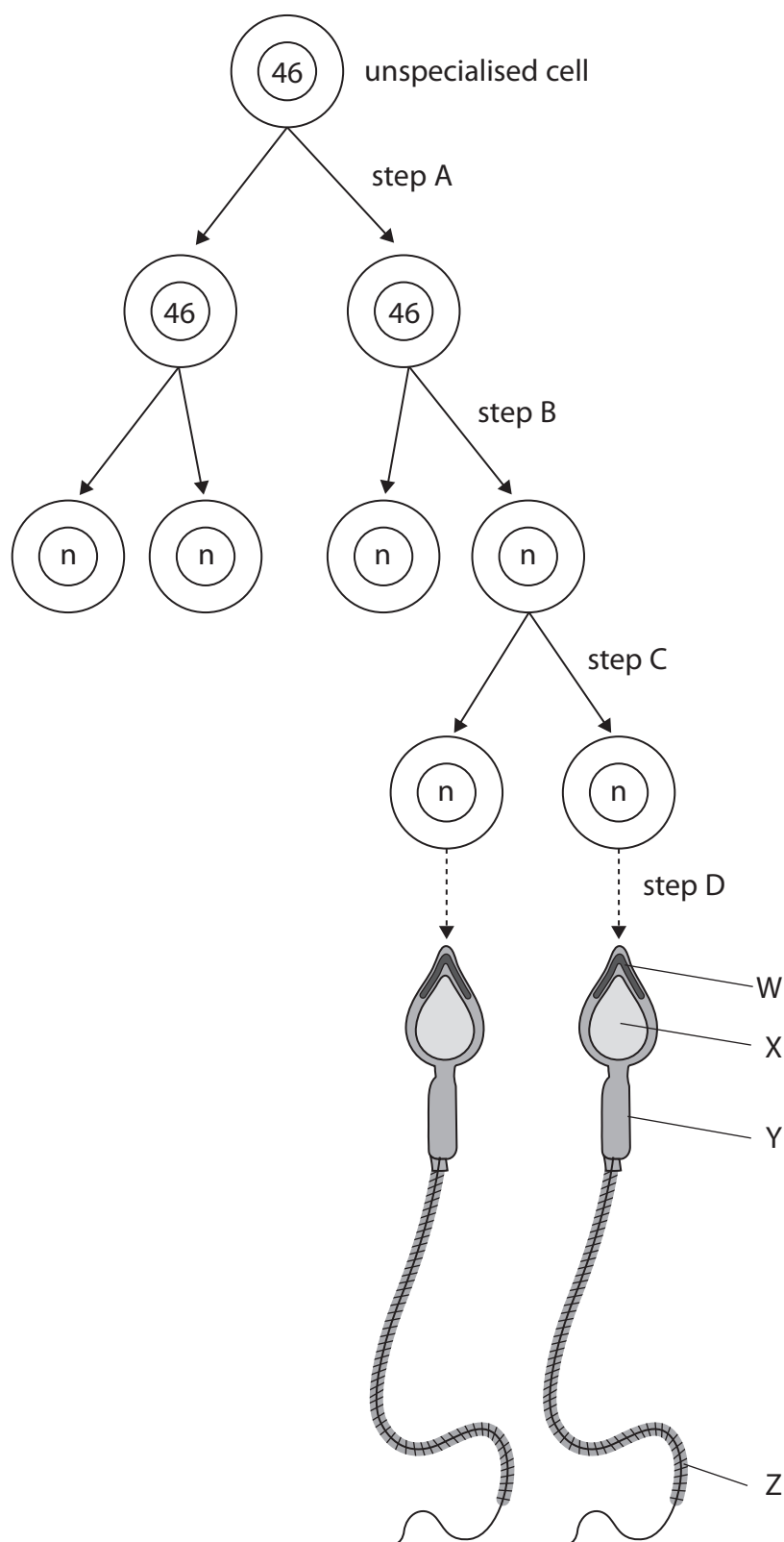


- 6 The diagram shows some of the steps involved in the production of human sperm cells from an un specialised cell. This process involves both mitosis and meiosis.



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(c) The nucleus of a sperm cell fuses with the nucleus of an egg cell to form a zygote. The zygote develops into a blastocyst.

(i) Draw a diagram to show the arrangement of the cells of a blastocyst.

(2)

(ii) The blastocyst contains some pluripotent stem cells.

Which of the following describes these pluripotent stem cells?

(1)

- ☐ **A** adult stem cells that can give rise to most types of cell in the body, including totipotent cells
- ☐ **B** adult stem cells that can give rise to most types of cell in the body, excluding totipotent cells
- ☐ **C** embryonic stem cells that can give rise to most types of cell in the body, including totipotent cells
- ☐ **D** embryonic stem cells that can give rise to most types of cell in the body, excluding totipotent cells



(d) The photograph shows a Mexican salamander.



(Source: © Imagebroker/Alamy Stock Photo)

Salamanders can regenerate damaged limbs and organs throughout their lives.

An ethics committee has approved the use of stem cells taken from salamanders.

Suggest why the use of stem cells taken from salamander embryos was approved.

(2)

**(Total for Question 6 = 13 marks)**